

Knowva ERD

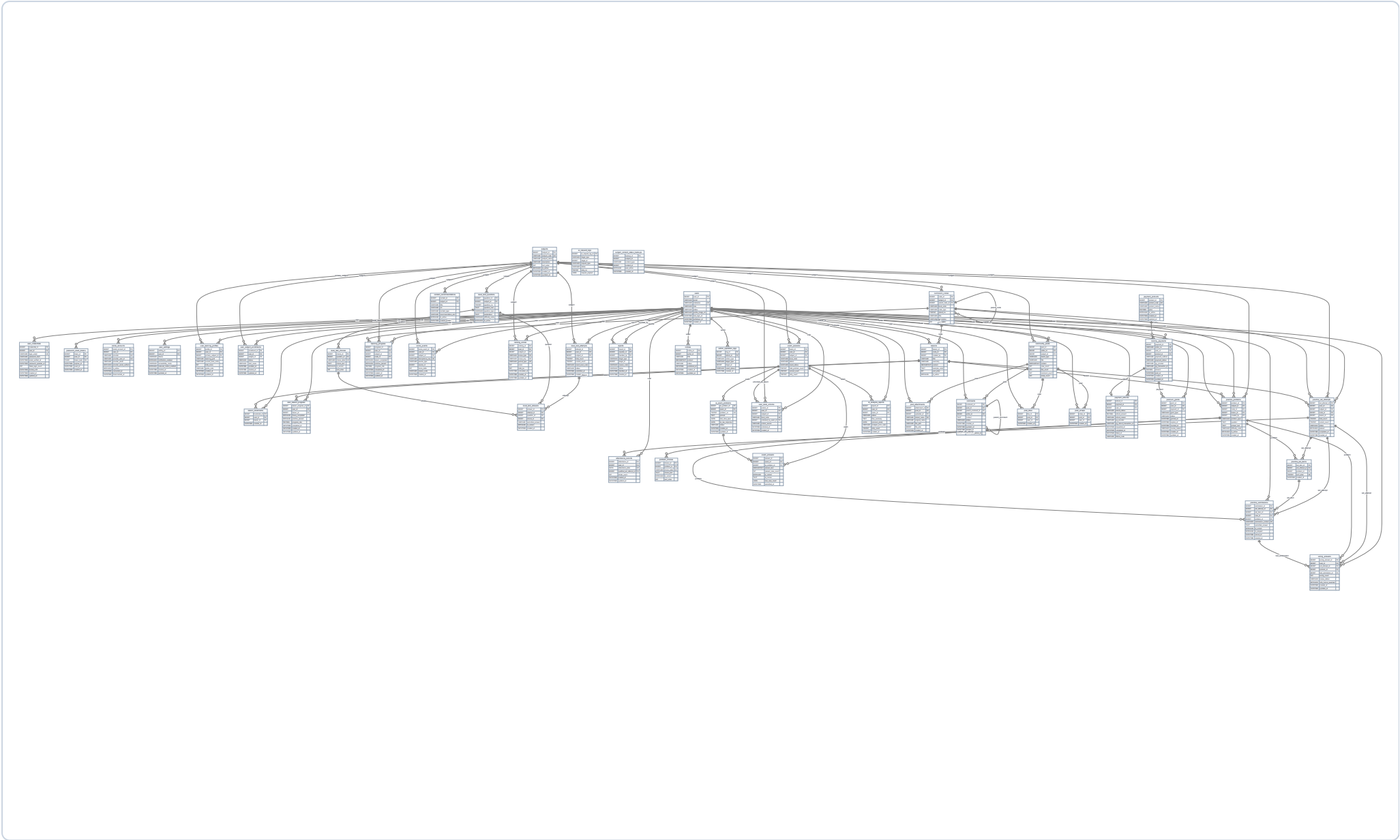
Source: Notion DB 명세 v2.4 Mermaid ERD. 관리자 댓글 삭제 주체와 비밀번호 재설정 token hash, 만료, 1회 사용 계약을 현재 DDL과 동기화한 HTML 원본.

Version v2.4	Tables 46	Foreign Keys 83	DBMS MySQL 8
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Table	PK	FK Count
users	BIGINT user_id PK	0
subjects	BIGINT subject_id PK	0
curriculum_nodes	BIGINT node_id PK	2
payment_products	BIGINT product_id PK	0
password_reset_tokens	BIGINT token_id PK	1
user_credentials	BIGINT credential_id PK	1
social_accounts	BIGINT social_account_id PK	1
user_settings	BIGINT setting_id PK	1
user_learning_profiles	BIGINT profile_id PK	2
user_subject_enrollments	BIGINT enrollment_id PK	2
lessons	BIGINT lesson_id PK	2
lesson_bookmarks	BIGINT bookmark_id PK	2
user_lesson_progress	BIGINT lesson_progress_id PK	2
level_test_questions	BIGINT question_id PK	1
level_test_choices	BIGINT choice_id PK	1
level_test_attempts	BIGINT attempt_id PK	2
level_test_answers	BIGINT answer_id PK	3
learning_progress	BIGINT progress_id PK	3
practice_set_attempts	BIGINT set_attempt_id PK	4
attendance_records	BIGINT attendance_id PK	2
practice_problems	BIGINT problem_id PK	4
problem_choices	BIGINT choice_id PK	1
practice_set_items	BIGINT set_item_id PK	2
practice_submissions	BIGINT submission_id PK	4
wrong_answers	BIGINT wrong_answer_id PK	4
score_events	BIGINT score_event_id PK	2
ranking_scores	BIGINT ranking_id PK	2
exam_sessions	BIGINT exam_id PK	2

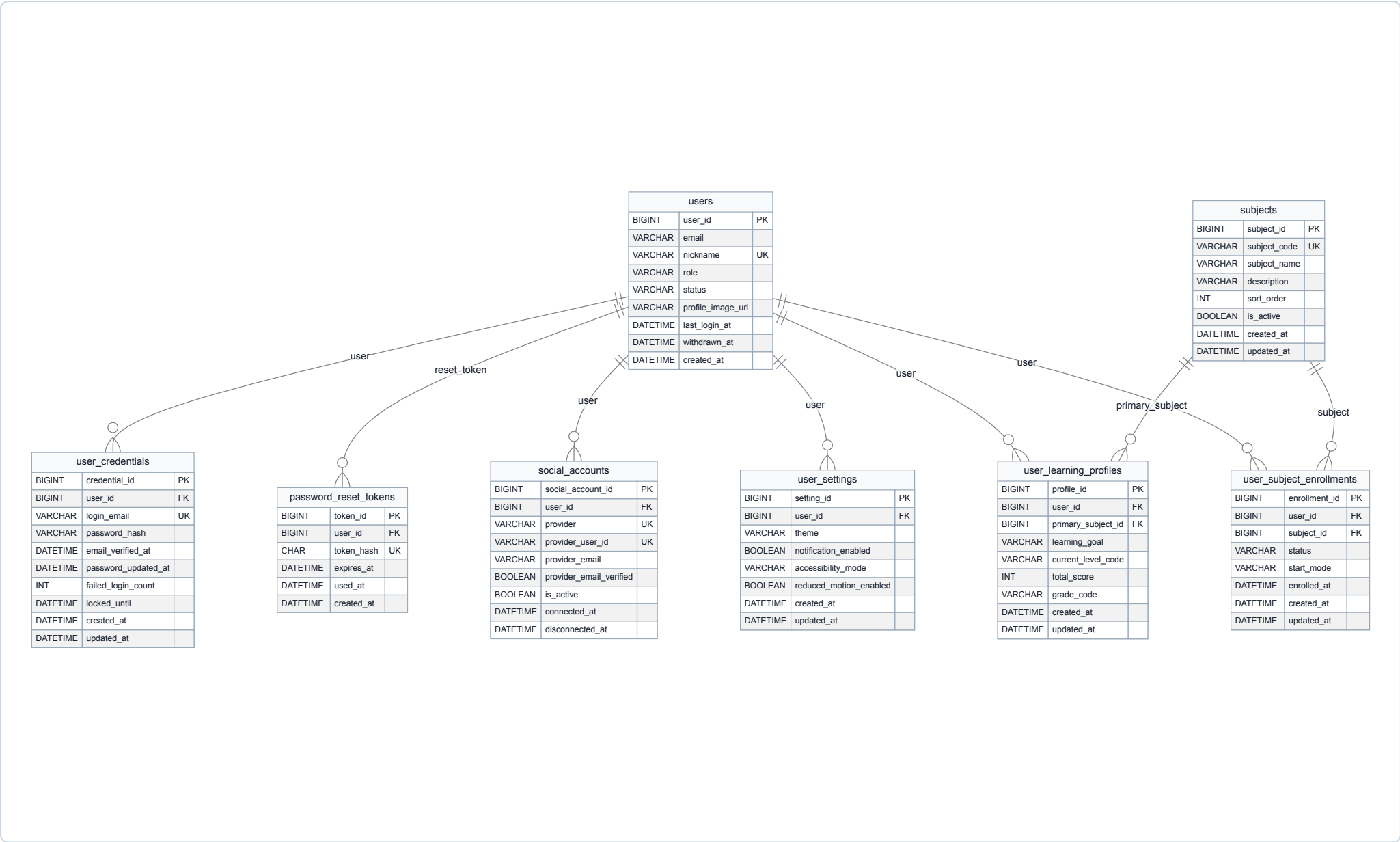
Table	PK	FK Count
user_level_unlocks	BIGINT unlock_id PK	3
ai_exam_problems	BIGINT ai_problem_id PK	1
exam_answers	BIGINT answer_id PK	2
ai_analysis_reports	BIGINT report_id PK	2
ai_request_logs	BIGINT ai_request_log_id PK	0
community_posts	BIGINT post_id PK	2
post_attachments	BIGINT attachment_id PK	2
comments	BIGINT comment_id PK	3
post_likes	BIGINT like_id PK	2
post_scraps	BIGINT scrap_id PK	2
reports	BIGINT report_id PK	2
dummy_payments	BIGINT payment_id PK	2
payment_refunds	BIGINT refund_id PK	2
premium_grants	BIGINT grant_id PK	2
notices	BIGINT notice_id PK	1
admin_operation_logs	BIGINT log_id PK	1
subject_content_status_backups	BIGINT backup_id PK	0
content_recommendations	BIGINT content_id PK	1

전체 ERD



전체 구조 확인용 축소 ERD. 상세 컬럼과 관계는 이후 섹션별 ERD에서 확인.

회원 / 인증 / 설정



회원 / 인증 / 설정 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
user_credentials	user_id	users	user_id
password_reset_tokens	user_id	users	user_id
social_accounts	user_id	users	user_id
user_settings	user_id	users	user_id
user_learning_profiles	user_id	users	user_id
user_learning_profiles	primary_subject_id	subjects	subject_id
user_subject_enrollments	user_id	users	user_id
user_subject_enrollments	subject_id	subjects	subject_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL		NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_credentials			
Column	Type	Key	NN
credential_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
login_email	VARCHAR(255) NOT NULL	UK	NN
password_hash	VARCHAR(255) NOT NULL		NN
email_verified_at	DATETIME NOT NULL		NN
password_updated_at	DATETIME NULL		NULL
failed_login_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
locked_until	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

password_reset_tokens			
Column	Type	Key	NN
token_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
token_hash	CHAR(64) NOT NULL	UK	NN
expires_at	DATETIME NOT NULL	IDX	NN
used_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

social_accounts

Column	Type	Key	NN
social_account_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
provider	VARCHAR(30) NOT NULL	UK	NN
provider_user_id	VARCHAR(191) NOT NULL	UK	NN
provider_email	VARCHAR(255) NULL		NULL
provider_email_verified	TINYINT(1) NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
connected_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
disconnected_at	DATETIME NULL		NULL

user_settings

Column	Type	Key	NN
setting_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
theme	VARCHAR(30) NOT NULL DEFAULT 'SYSTEM'		NN
notification_enabled	TINYINT(1) NOT NULL DEFAULT 1		NN
accessibility_mode	VARCHAR(30) NULL		NULL
reduced_motion_enabled	TINYINT(1) NOT NULL DEFAULT 0		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_learning_profiles

Column	Type	Key	NN
profile_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
primary_subject_id	BIGINT UNSIGNED NULL	FK	NULL
learning_goal	VARCHAR(500) NULL		NULL
current_level_code	VARCHAR(30) NOT NULL DEFAULT 'BRONZE'		NN
total_score	INT UNSIGNED NOT NULL DEFAULT 0		NN
grade_code	VARCHAR(30) NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

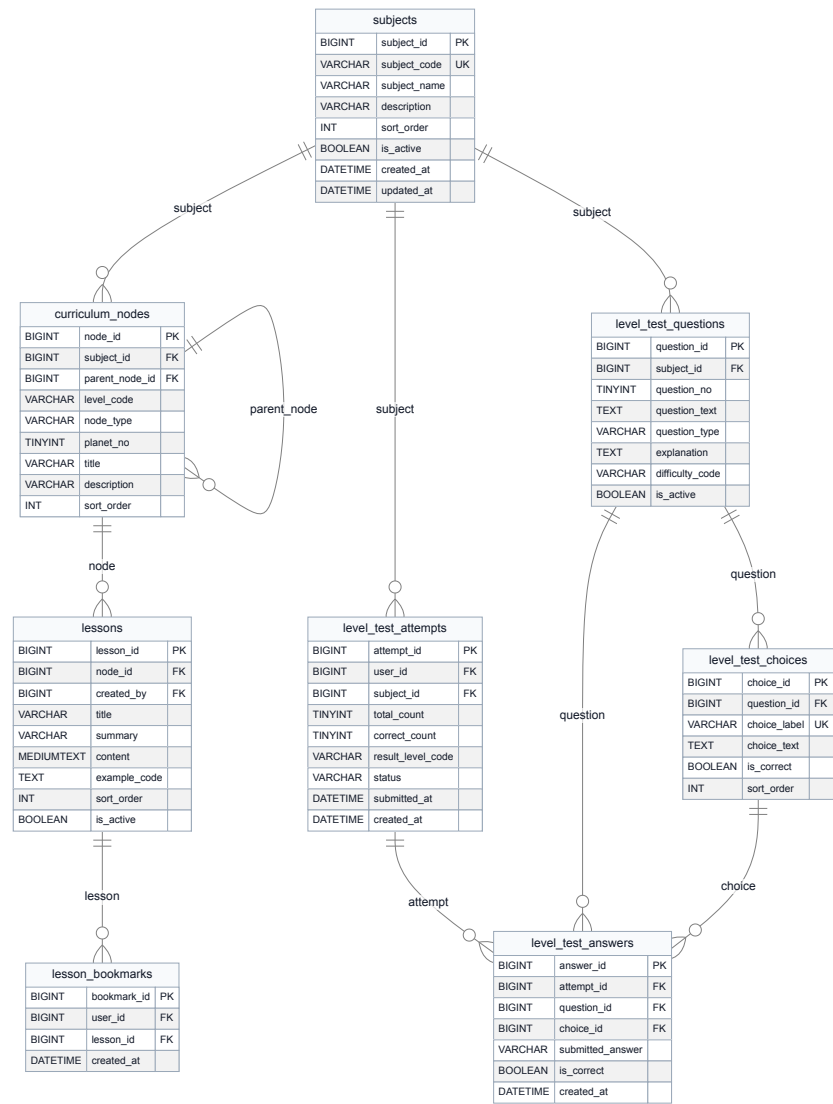
user_subject_enrollments

Column	Type	Key	NN
enrollment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'	IDX	NN
start_mode	VARCHAR(30) NOT NULL DEFAULT 'BASIC'		NN
enrolled_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects

Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

학습 Master / 레벨 테스트



학습 Master / 레벨 테스트 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
curriculum_nodes	subject_id	subjects	subject_id
curriculum_nodes	parent_node_id	curriculum_nodes	node_id
lessons	node_id	curriculum_nodes	node_id
lesson_bookmarks	lesson_id	lessons	lesson_id
level_test_questions	subject_id	subjects	subject_id
level_test_choices	question_id	level_test_questions	question_id
level_test_attempts	subject_id	subjects	subject_id
level_test_answers	attempt_id	level_test_attempts	attempt_id
level_test_answers	question_id	level_test_questions	question_id
level_test_answers	choice_id	level_test_choices	choice_id

subjects			
Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

curriculum_nodes			
Column	Type	Key	NN
node_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
parent_node_id	BIGINT UNSIGNED NULL	FK	NULL
level_code	VARCHAR(30) NOT NULL		NN
node_type	VARCHAR(30) NOT NULL DEFAULT 'PLANET'		NN
planet_no	TINYINT UNSIGNED NULL		NULL
title	VARCHAR(150) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
gate_condition	VARCHAR(255) NULL		NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

lessons			
Column	Type	Key	NN
lesson_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK	NN
title	VARCHAR(150) NOT NULL		NN
summary	VARCHAR(500) NULL		NULL
content	MEDIUMTEXT NOT NULL		NN
example_code	TEXT NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_by	BIGINT UNSIGNED NULL	FK	NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

lesson_bookmarks

Column	Type	Key	NN
bookmark_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
lesson_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

level_test_questions

Column	Type	Key	NN
question_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
question_no	TINYINT UNSIGNED NOT NULL		NN
question_text	TEXT NOT NULL		NN
question_type	VARCHAR(30) NOT NULL DEFAULT 'MULTIPLE_CHOICE'		NN
explanation	TEXT NULL		NU LL
difficulty_code	VARCHAR(30) NOT NULL DEFAULT 'BRONZE'		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

level_test_choices

Column	Type	Key	NN
choice_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
question_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
choice_label	VARCHAR(10) NOT NULL	UK	NN
choice_text	TEXT NOT NULL		NN
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN

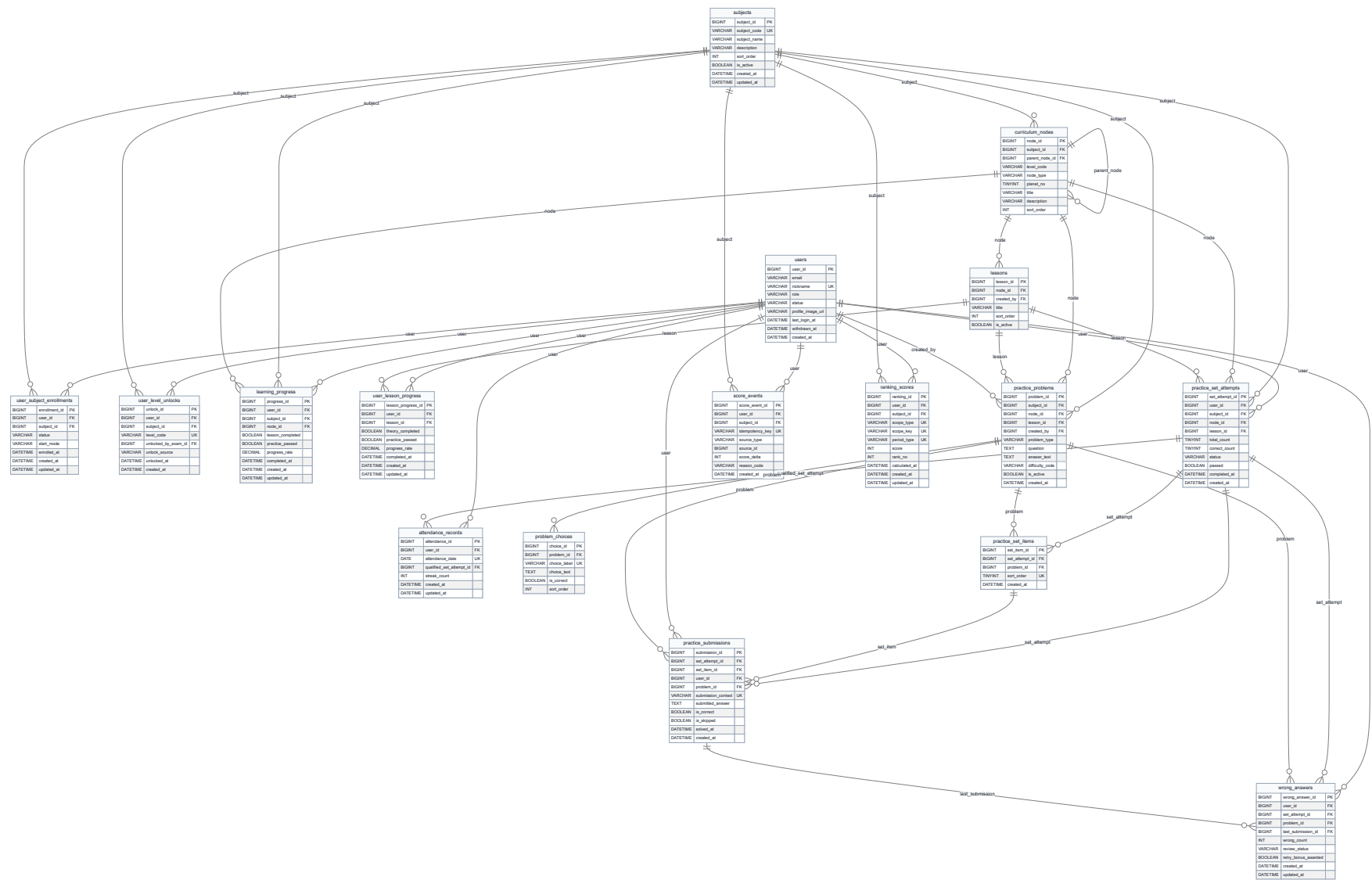
level_test_attempts

Column	Type	Key	NN
attempt_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
total_count	TINYINT UNSIGNED NOT NULL DEFAULT 8		NN
correct_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
result_level_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'SUBMITTED'		NN
submitted_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

level_test_answers

Column	Type	Key	NN
answer_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
attempt_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
question_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
choice_id	BIGINT UNSIGNED NULL	FK	NU LL
submitted_answer	VARCHAR(500) NULL		NU LL
is_correct	TINYINT(1) NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

학습 기록 / 일반 문제 / 점수



Child Table	FK Column	Parent Table	Parent PK
curriculum_nodes	subject_id	subjects	subject_id
curriculum_nodes	parent_node_id	curriculum_nodes	node_id
user_subject_enrollments	user_id	users	user_id
user_subject_enrollments	subject_id	subjects	subject_id
user_level_unlocks	user_id	users	user_id
user_level_unlocks	subject_id	subjects	subject_id
learning_progress	user_id	users	user_id
learning_progress	subject_id	subjects	subject_id
learning_progress	node_id	curriculum_nodes	node_id
user_lesson_progress	user_id	users	user_id
user_lesson_progress	lesson_id	lessons	lesson_id
attendance_records	user_id	users	user_id
attendance_records	qualified_set_attempt_id	practice_set_attempts	set_attempt_id
practice_set_attempts	user_id	users	user_id
practice_set_attempts	subject_id	subjects	subject_id
practice_set_attempts	node_id	curriculum_nodes	node_id
practice_set_attempts	lesson_id	lessons	lesson_id
practice_problems	subject_id	subjects	subject_id
practice_problems	node_id	curriculum_nodes	node_id
practice_problems	lesson_id	lessons	lesson_id
practice_problems	created_by	users	user_id
problem_choices	problem_id	practice_problems	problem_id
practice_set_items	set_attempt_id	practice_set_attempts	set_attempt_id
practice_set_items	problem_id	practice_problems	problem_id
practice_submissions	set_attempt_id	practice_set_attempts	set_attempt_id
practice_submissions	set_item_id	practice_set_items	set_item_id
practice_submissions	user_id	users	user_id
practice_submissions	problem_id	practice_problems	problem_id
wrong_answers	user_id	users	user_id
wrong_answers	set_attempt_id	practice_set_attempts	set_attempt_id

Child Table	FK Column	Parent Table	Parent PK
wrong_answers	problem_id	practice_problems	problem_id
wrong_answers	last_submission_id	practice_submissions	submission_id
score_events	user_id	users	user_id
score_events	subject_id	subjects	subject_id
ranking_scores	user_id	users	user_id
ranking_scores	subject_id	subjects	subject_id

users

Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects

Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

curriculum_nodes

Column	Type	Key	NN
node_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
parent_node_id	BIGINT UNSIGNED NULL	FK	NULL
level_code	VARCHAR(30) NOT NULL		NN
node_type	VARCHAR(30) NOT NULL DEFAULT 'PLANET'		NN
planet_no	TINYINT UNSIGNED NULL		NULL
title	VARCHAR(150) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
gate_condition	VARCHAR(255) NULL		NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_subject_enrollments

Column	Type	Key	NN
enrollment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'	IDX	NN
start_mode	VARCHAR(30) NOT NULL DEFAULT 'BASIC'		NN
enrolled_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_level_unlocks

Column	Type	Key	NN
unlock_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
level_code	VARCHAR(30) NOT NULL	UK	NN
unlock_source	VARCHAR(30) NOT NULL		NN
unlocked_by_exam_id	BIGINT UNSIGNED NULL	FK	NULL
unlocked_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

learning_progress

Column	Type	Key	NN
progress_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
lesson_completed	TINYINT(1) NOT NULL DEFAULT 0		NN
practice_passed	TINYINT(1) NOT NULL DEFAULT 0		NN
progress_rate	DECIMAL(5,2) NOT NULL DEFAULT 0.00		NN
completed_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_lesson_progress

Column	Type	Key	NN
lesson_progress_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
lesson_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
theory_completed	TINYINT(1) NOT NULL DEFAULT 0		NN
practice_passed	TINYINT(1) NOT NULL DEFAULT 0		NN
progress_rate	DECIMAL(5,2) NOT NULL DEFAULT 0.00		NN
completed_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

attendance_records

Column	Type	Key	NN
attendance_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
attendance_date	DATE NOT NULL	UK	NN
streak_count	INT UNSIGNED NOT NULL DEFAULT 1		NN
qualified_set_attempt_id	BIGINT UNSIGNED NULL	FK UK	NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

practice_set_attempts

Column	Type	Key	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK	NN
lesson_id	BIGINT UNSIGNED NULL	FK	NULL
total_count	TINYINT UNSIGNED NOT NULL DEFAULT 10		NN
correct_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
status	VARCHAR(30) NOT NULL DEFAULT 'IN_PROGRESS'		NN
passed	TINYINT(1) NOT NULL DEFAULT 0		NN
completed_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

practice_problems

Column	Type	Key	NN
problem_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK	NN
lesson_id	BIGINT UNSIGNED NULL	FK	NULL
problem_type	VARCHAR(30) NOT NULL DEFAULT 'MULTIPLE_CHOICE'		NN
question	TEXT NOT NULL		NN
answer_text	TEXT NULL		NULL
difficulty_code	VARCHAR(30) NOT NULL DEFAULT 'BRONZE'		NN
created_by	BIGINT UNSIGNED NULL	FK	NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

problem_choices

Column	Type	Key	NN
choice_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
choice_label	VARCHAR(10) NOT NULL	UK	NN
choice_text	TEXT NOT NULL		NN
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN

practice_set_items

Column	Type	Key	NN
set_item_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
sort_order	TINYINT UNSIGNED NOT NULL	UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

practice_submissions

Column	Type	Key	NN
submission_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
set_item_id	BIGINT UNSIGNED NULL	FK	NULL
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
submission_context	VARCHAR(30) NOT NULL DEFAULT 'PRACTICE_SET'	UK	NN
submitted_answer	TEXT NULL		NULL
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
is_skipped	TINYINT(1) NOT NULL DEFAULT 0		NN
solved_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

wrong_answers

Column	Type	Key	NN
wrong_answer_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL	FK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
last_submission_id	BIGINT UNSIGNED NULL	FK	NULL
wrong_count	INT UNSIGNED NOT NULL DEFAULT 1		NN
review_status	VARCHAR(30) NOT NULL DEFAULT 'OPEN'		NN
retry_bonus_awarded	TINYINT(1) NOT NULL DEFAULT 0		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

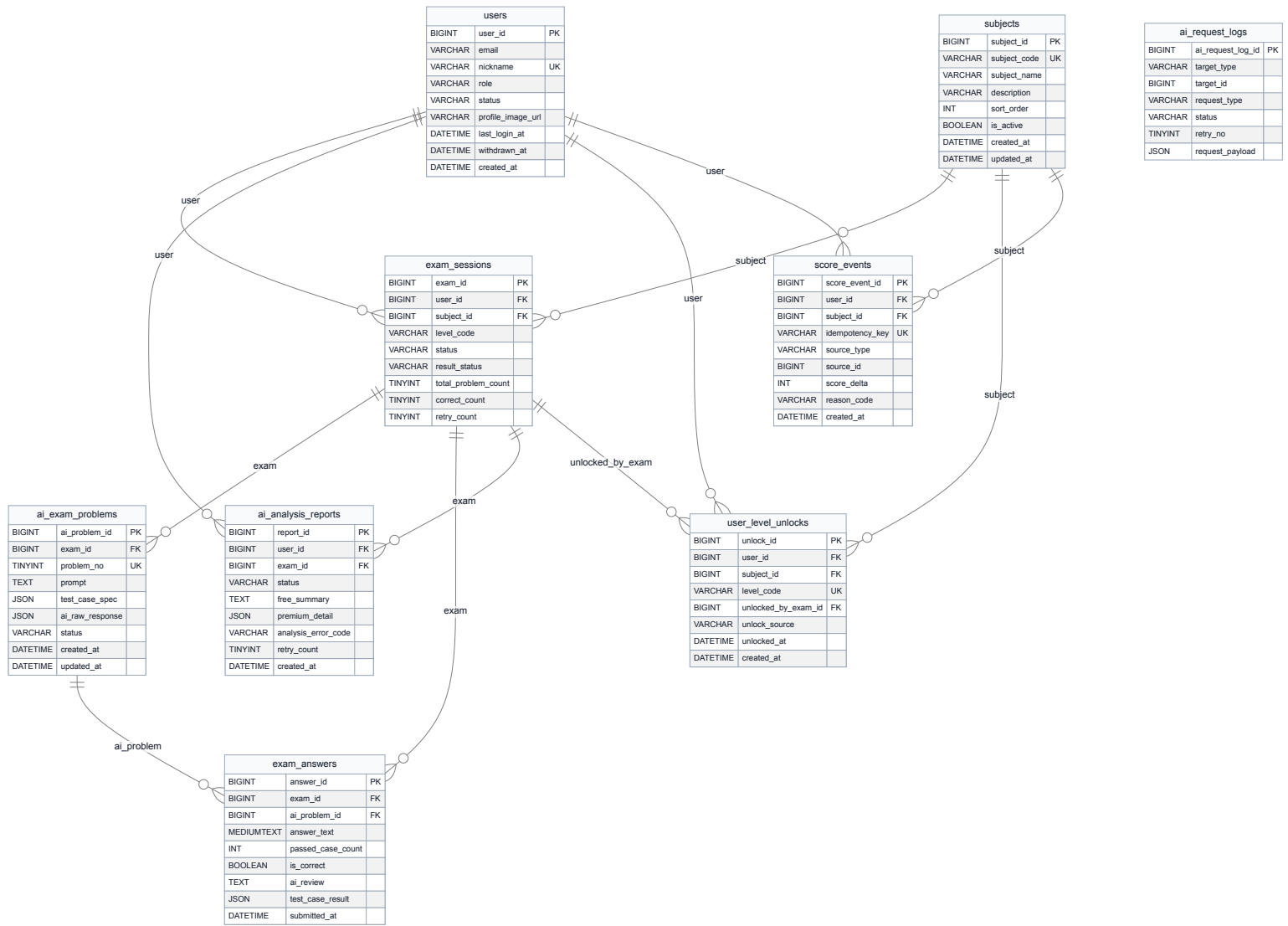
score_events

Column	Type	Key	NN
score_event_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
source_type	VARCHAR(30) NOT NULL		NN
source_id	BIGINT UNSIGNED NOT NULL		NN
score_delta	INT NOT NULL		NN
reason_code	VARCHAR(30) NOT NULL		NN
idempotency_key	VARCHAR(191) NOT NULL	UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

ranking_scores

Column	Type	Key	NN
ranking_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NULL	FK	NU LL
scope_type	VARCHAR(30) NOT NULL	UK	NN
scope_key	VARCHAR(100) NOT NULL	UK	NN
period_type	VARCHAR(30) NOT NULL	UK	NN
period_key	VARCHAR(30) NOT NULL	UK	NN
score	INT UNSIGNED NOT NULL DEFAULT 0		NN
rank_no	INT UNSIGNED NULL		NU LL
calculated_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

AI 시험 / AI 분석



AI 시험 / AI 분석 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
exam_sessions	user_id	users	user_id
exam_sessions	subject_id	subjects	subject_id
ai_exam_problems	exam_id	exam_sessions	exam_id
exam_answers	exam_id	exam_sessions	exam_id
exam_answers	ai_problem_id	ai_exam_problems	ai_problem_id
ai_analysis_reports	user_id	users	user_id
ai_analysis_reports	exam_id	exam_sessions	exam_id
user_level_unlocks	user_id	users	user_id
user_level_unlocks	subject_id	subjects	subject_id
user_level_unlocks	unlocked_by_exam_id	exam_sessions	exam_id
score_events	user_id	users	user_id
score_events	subject_id	subjects	subject_id

users

Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects

Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

exam_sessions

Column	Type	Key	NN
exam_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
level_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'CREATED'		NN
result_status	VARCHAR(30) NULL		NULL
total_problem_count	TINYINT UNSIGNED NOT NULL DEFAULT 3		NN
correct_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
retry_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
started_at	DATETIME NULL		NULL
submitted_at	DATETIME NULL		NULL
graded_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

ai_exam_problems

Column	Type	Key	NN
ai_problem_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
exam_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
problem_no	TINYINT UNSIGNED NOT NULL	UK	NN
prompt	TEXT NOT NULL		NN
test_case_spec	JSON NULL		NULL
ai_raw_response	JSON NULL		NULL
status	VARCHAR(30) NOT NULL DEFAULT 'GENERATED'		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

exam_answers

Column	Type	Key	NN
answer_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
exam_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
ai_problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
answer_text	MEDIUMTEXT NOT NULL		NN
passed_case_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
ai_review	TEXT NULL		NULL
test_case_result	JSON NULL		NULL
submitted_at	DATETIME NULL		NULL
graded_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

ai_analysis_reports

Column	Type	Key	NN
report_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
exam_id	BIGINT UNSIGNED NOT NULL	FK	NN
status	VARCHAR(30) NOT NULL DEFAULT 'PENDING'		NN
free_summary	TEXT NULL		NULL
premium_detail	JSON NULL		NULL
analysis_error_code	VARCHAR(100) NULL		NULL
retry_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

ai_request_logs

Column	Type	Key	NN
ai_request_log_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NOT NULL		NN
request_type	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL		NN
retry_no	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
request_payload	JSON NULL		NULL
response_payload	JSON NULL		NULL
error_code	VARCHAR(100) NULL		NULL
error_message	VARCHAR(1000) NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

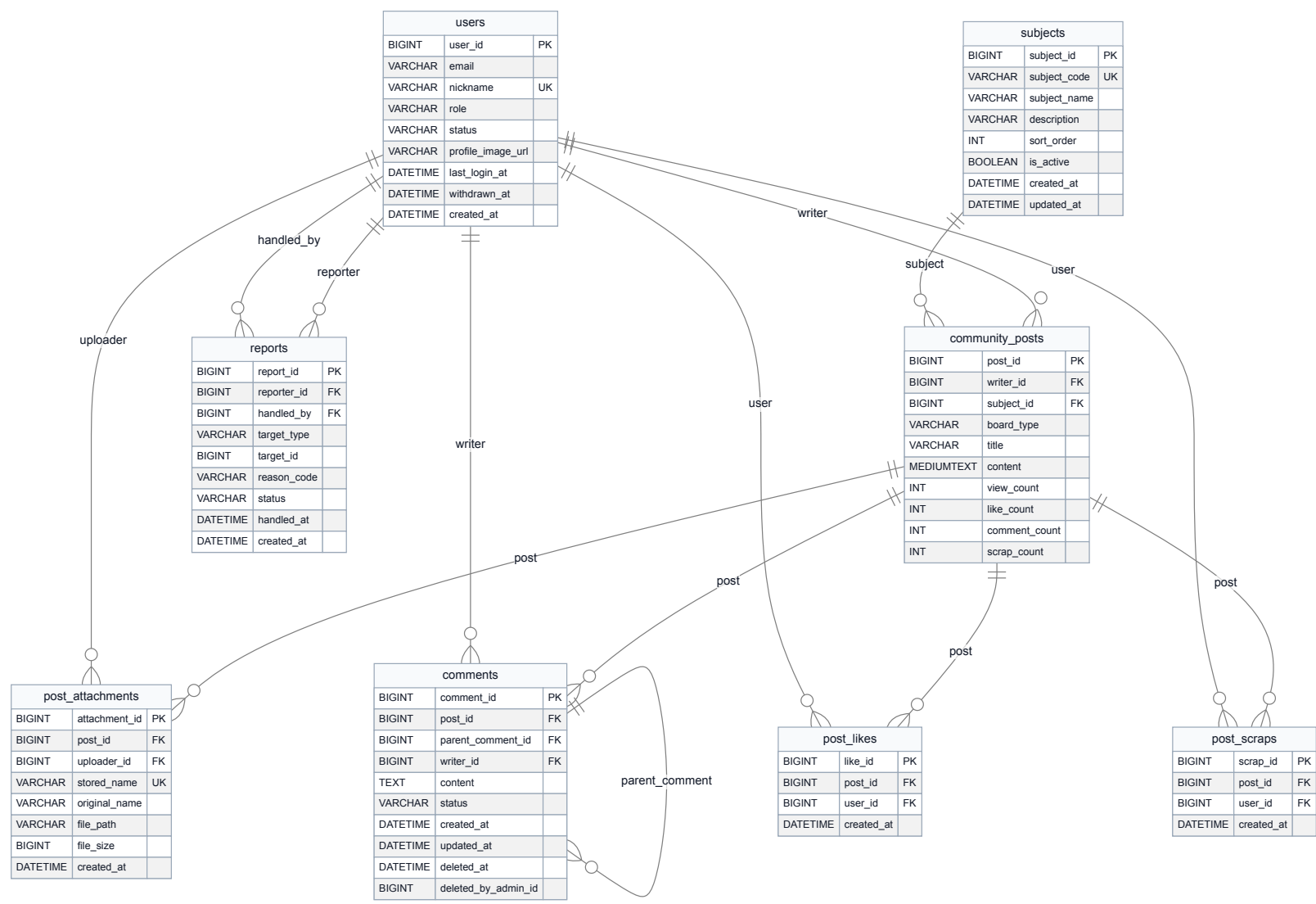
user_level_unlocks

Column	Type	Key	NN
unlock_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
level_code	VARCHAR(30) NOT NULL	UK	NN
unlock_source	VARCHAR(30) NOT NULL		NN
unlocked_by_exam_id	BIGINT UNSIGNED NULL	FK	NULL
unlocked_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

score_events

Column	Type	Key	NN
score_event_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
source_type	VARCHAR(30) NOT NULL		NN
source_id	BIGINT UNSIGNED NOT NULL		NN
score_delta	INT NOT NULL		NN
reason_code	VARCHAR(30) NOT NULL		NN
idempotency_key	VARCHAR(191) NOT NULL	UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

커뮤니티



커뮤니티 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
community_posts	writer_id	users	user_id
community_posts	subject_id	subjects	subject_id
post_attachments	post_id	community_posts	post_id
post_attachments	uploader_id	users	user_id
comments	post_id	community_posts	post_id
comments	parent_comment_id	comments	comment_id
comments	writer_id	users	user_id
post_likes	post_id	community_posts	post_id
post_likes	user_id	users	user_id
post_scraps	post_id	community_posts	post_id
post_scraps	user_id	users	user_id
reports	reporter_id	users	user_id
reports	handled_by	users	user_id

users

Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects

Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

community_posts

Column	Type	Key	NN
post_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
writer_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
board_type	VARCHAR(30) NOT NULL		NN
title	VARCHAR(200) NOT NULL		NN
content	MEDIUMTEXT NOT NULL		NN
view_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
like_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
comment_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
scrap_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL
deleted_at	DATETIME NULL		NULL

post_attachments

Column	Type	Key	NN
attachment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK	NN
uploader_id	BIGINT UNSIGNED NOT NULL	FK	NN
original_name	VARCHAR(255) NOT NULL		NN
stored_name	VARCHAR(255) NOT NULL	UK	NN
file_path	VARCHAR(500) NOT NULL		NN
file_size	BIGINT UNSIGNED NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

post_scraps

Column	Type	Key	NN
scrap_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

comments

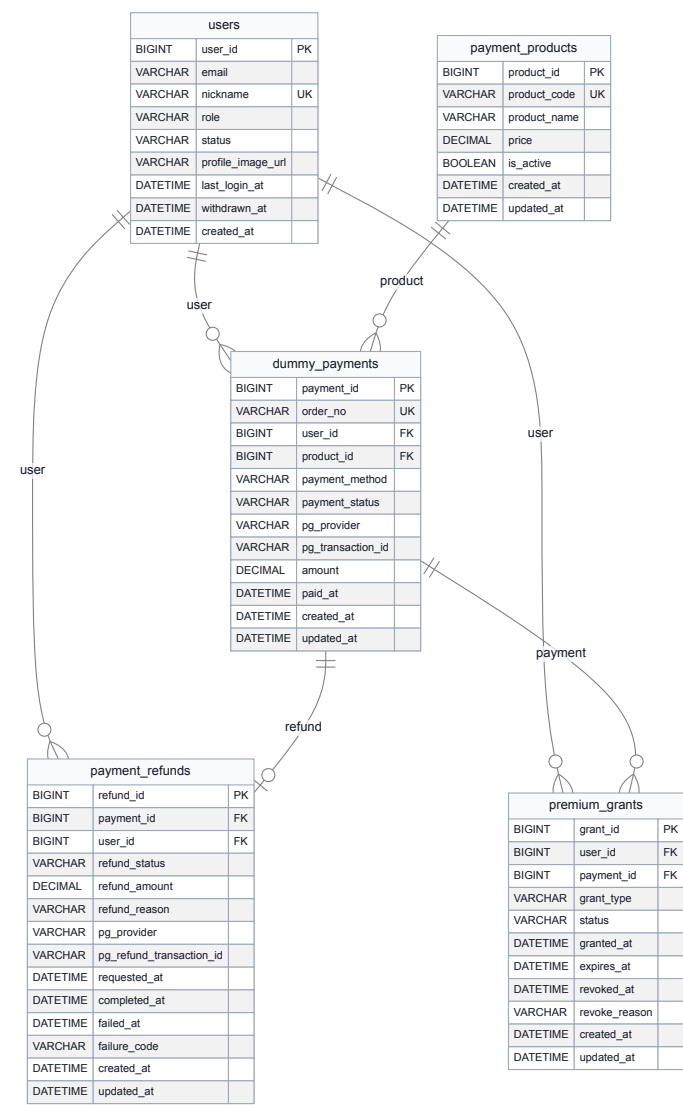
Column	Type	Key	NN
comment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK	NN
parent_comment_id	BIGINT UNSIGNED NULL	FK	NU LL
writer_id	BIGINT UNSIGNED NOT NULL	FK	NN
content	TEXT NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL
deleted_at	DATETIME NULL		NU LL
deleted_by_admin_id	BIGINT NULL		NU LL

reports

Column	Type	Key	NN
report_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NOT NULL		NN
reporter_id	BIGINT UNSIGNED NOT NULL	FK	NN
reason_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'RECEIVED'		NN
handled_by	BIGINT UNSIGNED NULL	FK	NU LL
handled_at	DATETIME NULL		NU LL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

post_likes

Column	Type	Key	NN
like_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN



결제 / Premium - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
dummy_payments	user_id	users	user_id
dummy_payments	product_id	payment_products	product_id
payment_refunds	payment_id	dummy_payments	payment_id
payment_refunds	user_id	users	user_id
premium_grants	user_id	users	user_id
premium_grants	payment_id	dummy_payments	payment_id

users

Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

payment_products

Column	Type	Key	NN
product_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
product_code	VARCHAR(30) NOT NULL	UK	NN
product_name	VARCHAR(100) NOT NULL		NN
price	DECIMAL(10,2) NOT NULL DEFAULT 0.00		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

dummy_payments

Column	Type	Key	NN
payment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
order_no	VARCHAR(100) NOT NULL	UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
product_id	BIGINT UNSIGNED NOT NULL	FK	NN
payment_method	VARCHAR(30) NOT NULL DEFAULT 'DUMMY'		NN
payment_status	VARCHAR(30) NOT NULL DEFAULT 'PENDING'		NN
pg_provider	VARCHAR(30) NULL	UK (composite)	NULL
pg_transaction_id	VARCHAR(200) NULL	UK (composite)	NULL
amount	DECIMAL(10,2) NOT NULL DEFAULT 0.00		NN
paid_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

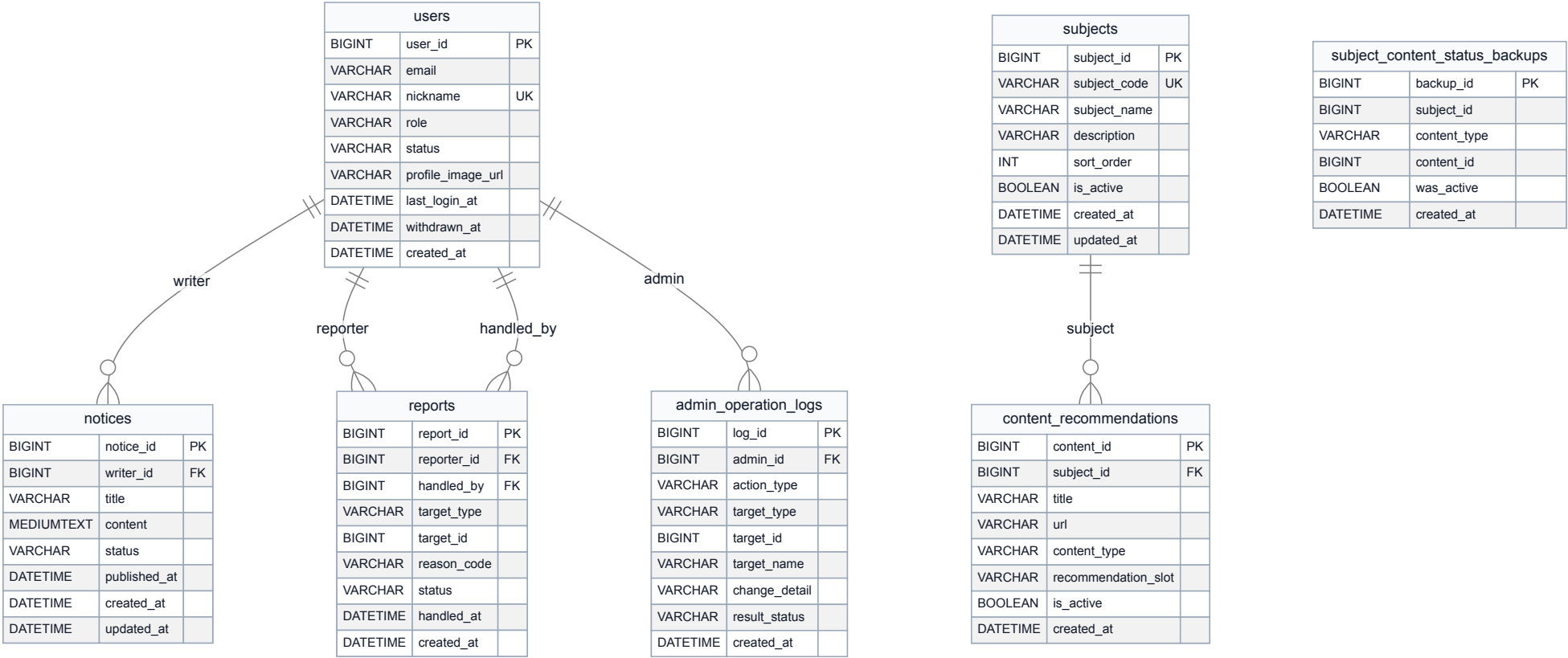
payment_refunds

Column	Type	Key	NN
refund_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
payment_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
refund_status	VARCHAR(30) NOT NULL DEFAULT 'PENDING'		NN
refund_amount	DECIMAL(10,2) NOT NULL		NN
refund_reason	VARCHAR(200) NOT NULL		NN
pg_provider	VARCHAR(30) NOT NULL	IDX (composite)	NN
pg_refund_transaction_id	VARCHAR(200) NULL	IDX (composite)	NULL
requested_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
completed_at	DATETIME NULL		NULL
failed_at	DATETIME NULL		NULL
failure_code	VARCHAR(100) NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

premium_grants

Column	Type	Key	NN
grant_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
payment_id	BIGINT UNSIGNED NULL	FK UK	NULL
grant_type	VARCHAR(30) NOT NULL DEFAULT 'LIFETIME'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
granted_at	DATETIME NOT NULL		NN
expires_at	DATETIME NULL		NULL
revoked_at	DATETIME NULL		NULL
revoke_reason	VARCHAR(200) NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

관리자 / 공지 / 확장 콘텐츠



Child Table	FK Column	Parent Table	Parent PK
notices	writer_id	users	user_id
reports	reporter_id	users	user_id
reports	handled_by	users	user_id
admin_operation_logs	admin_id	users	user_id
content_recommendations	subject_id	subjects	subject_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects			
Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

notices			
Column	Type	Key	NN
notice_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
writer_id	BIGINT UNSIGNED NOT NULL	FK	NN
title	VARCHAR(200) NOT NULL		NN
content	MEDIUMTEXT NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'DRAFT'		NN
published_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

reports

Column	Type	Key	NN
report_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NOT NULL		NN
reporter_id	BIGINT UNSIGNED NOT NULL	FK	NN
reason_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'RECEIVED'		NN
handled_by	BIGINT UNSIGNED NULL	FK	NULL
handled_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

content_recommendations

Column	Type	Key	NN
content_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
title	VARCHAR(200) NOT NULL		NN
url	VARCHAR(500) NOT NULL		NN
content_type	VARCHAR(30) NOT NULL		NN
recommendation_slot	VARCHAR(30) NOT NULL		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

admin_operation_logs

Column	Type	Key	NN
log_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
admin_id	BIGINT UNSIGNED NOT NULL	FK	NN
action_type	VARCHAR(50) NOT NULL		NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NULL		NULL
target_name	VARCHAR(255) NULL		NULL
change_detail	VARCHAR(500) NULL		NULL
result_status	VARCHAR(30) NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

subject_content_status_backups

Column	Type	Key	NN
backup_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	UK (composite)	NN
content_type	VARCHAR(30) NOT NULL	UK (composite)	NN
content_id	BIGINT UNSIGNED NOT NULL	UK (composite)	NN
was_active	TINYINT(1) NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

v2.4 변경 요약

관리자 댓글 삭제 주체

- `comments.deleted_by_admin_id`: 관리자 삭제 처리 시 처리자 ID를 저장한다. 사용자 직접 삭제와 관리자 숨김은 **NULL**로 유지한다.
- 댓글 상태가 **DELETED**일 때 이 값이 있으면 “관리자에 의해 삭제된 댓글입니다.”, 없으면 “삭제된 댓글입니다.”로 표시한다.
- 관리자 공개 복구 시 `deleted_at`과 `deleted_by_admin_id`를 함께 **NULL**로 초기화한다.

v2.3 변경 요약

비밀번호 재설정 token

- `password_reset_tokens`: 이메일/비밀번호 계정의 재설정 token hash, 만료 시각, 1회 사용 시각을 보관한다.
- `token_hash`: SHA-256 hash만 unique로 저장한다. raw token은 DB, sample data, application log에 저장하지 않는다.
- `expires_at`, `used_at`: 만료되지 않았고 사용되지 않은 token만 비밀번호 변경에 사용할 수 있다.

v2.2 변경 요약

Premium 환불과 권한 회수

- `payment_refunds`: 결제당 1건의 환불 요청, PG 성공·실패, 환불 PG 거래 식별자를 보존한다.
- `dummy_payments`: PG 환불 성공 시에만 원결제를 **REFUNDED**로 전환한다.
- `premium_grants`: 성공한 환불에서만 **REVOKED**, `revoked_at`, `revoke_reason`으로 권한 회수를 기록한다.
- `ai_analysis_reports`: `idx_ai_analysis_reports_refund_eligibility(user_id, status)`로 성공 Premium 상세 분석 사용 이력의 환불 제한 조회를 보장한다.

v2.1 변경 요약

과목 수강 상태와 관리자 이력 상세

- `user_subject_enrollments`: 사용자별 과목 수강 상태와 시작 방식을 저장한다. (`user_id`, `subject_id`) 복합 unique로 중복 수강을 막고, 학습 화면은 **ACTIVE** 과목만 조회한다.
- `admin_operation_logs.target_name`: 관리자 action 당시의 대상명을 보존한다. 이후 대상이 삭제되거나 이름이 바뀌어도 이력 상세는 유지된다.
- `admin_operation_logs.change_detail`: 실제 적용한 변경 내용을 보존해 운영 로그의 문장형 상세 표시를 지원한다.

v2.0 변경 요약

과목 콘텐츠 복원과 PG 전환 준비

- `subject_content_status_backups`: 과목 비활성화 직전의 CURRICULUM, LESSON, PROBLEM 활성 상태를 보존하고 재활성화 시 복원한다. (`subject_id`, `content_type`, `content_id`) 복합 unique로 중복 스냅샷을 차단한다.
- `dummy_payments`: 결제 시작 주문은 기본 PENDING 상태로 생성한다. `pg_provider`와 `pg_transaction_id`로 카카오페이 tid와 토스페이먼츠 paymentKey를 저장한다.
- `uk_dummy_payments_pg_transaction`: 동일 PG 안의 거래 식별자 중복을 차단하며, 기존 dummy 결제의 두 PG 컬럼은 NULL을 허용한다.

v1.9 변경 요약

커뮤니티 조회수 및 seed 실행 안전성

- `community_posts.view_count`: 게시글 상세 조회 시 증가하며 목록의 조회수 정렬과 화면 표시에 사용.
- `Knowva_sample_data.sql`: DB v1.9 필수 컬럼 preflight와 단일 transaction을 적용해 오래된 schema에서 부분 반영되는 문제를 차단.
- `Knowva_DDL.sql`: 기존 데이터를 보존하는 migration이 아니라 전체 재생성 script임을 명시.

v1.8 변경 요약

인증 수단별 독립 계정 정책

- `users.email`: UNIQUE 제거. 이메일은 연락처/표시 정보이며 계정 병합 기준으로 사용하지 않음.
- `user_credentials.login_email`: 이메일/비밀번호 로그인 식별자. 이메일 가입은 인증 완료 후 별도 계정으로 생성.
- `social_accounts.provider + provider_user_id`: Google/GitHub 소셜 로그인 식별자. provider 이메일이 기존 회원과 같아도 새 `users` 계정 생성.
- `provider`: google, github 소문자 값으로 통일.

v1.5 변경 요약

v1.4 AI 정책 반영

- `ai_exam_problems.test_case_spec`: ChatGPT API가 생성/보조한 테스트케이스 후보 저장
- `exam_answers.passed_case_count`, `test_case_result`: 정답/오답 판정은 테스트케이스 실행 결과로 처리
- `exam_answers.ai_review`: AI 해설/코드 리뷰 저장
- `ai_request_logs`: ChatGPT API 요청/응답/실패/재시도 payload 기록

구분	반영 내용
일반 문제 세트	<code>practice_set_attempts.subject_id</code> 추가. 문제 세트 <code>attempt</code> 에서 과목을 직접 추적.
오답 복습	<code>wrong_answers.set_attempt_id</code> 추가. 오답이 발생한 문제 세트 <code>attempt</code> 를 직접 추적.
로그인/권한	관리자는 사용자와 같은 로그인 화면을 사용하고, 로그인 후 <code>role</code> 에 따라 학습 또는 관리자 메인으로 이동.
AI Provider	로컬 LLM 미사용. ChatGPT API 호출/응답/실패/재시도 기록을 <code>ai_request_logs</code> 에 저장.
AI 코딩테스트	AI는 문제 생성, 테스트케이스 생성 보조, 해설 생성, 코드 리뷰 담당. 정답/오답은 <code>testcase</code> 실행 결과로 판정.
DDL 실행	전체 실행 시 <code>elearning schema</code> 의 <code>Knowva table</code> 을 <code>DROP TABLE IF EXISTS</code> 후 다시 생성.